Vescera at HalluScoring 2026: Privileged-Information Distillation with Contrastive Representation Alignment for Arabic Hallucination Detection

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Abstract

HalluScoring 2026 Subtask 2.2 (Hallucination Detection & Find the Truth) requires systems to judge whether an Arabic generated answer is hallucinated and to select the correct one among six candidates, without access to the reference answer that would make both judgments trivial. We treat that withheld reference as privileged information: a teacher is fine-tuned with it in the prompt, then never deployed, and its advantage is transferred to a blind student through two channels knowledge distillation over the teacher's output distribution, and a contrastive privileged-information (CPI) term that aligns the student's hidden representation with the teacher's gold-conditioned one. Our fine-tuned ALLaM-7B attains 84.62 combined score (Comb) on the official test set and 83.47 on the dev set.

1 Introduction

A fluent but factually wrong answer to an Arabic question is not a cosmetic failure but a trust-breaking one. **HalluScoring 2026 Subtask 2.2 (Hallucination Detection & Find the Truth)** poses hallucination detection and answer selection as one joint problem: given a question, a candidate answer, and six options, a system predicts a faithfulness label and the correct letter, scored by a weighted combination of label F1 and option accuracy. We adopt the task formulation and the HalluTruthQA-4K resource of HalluScoring 2026 Track 2 (Alansari et al., 2026; Bekhouche et al., 2026), which pairs each Arabic question with a model-generated response, a gold answer $a^\star$ (the verified, human-checked correct answer against which faithfulness is judged), and five distractors. Whether $a^\star$ is visible is what separates an easy version of this task from a hard one: exposing the gold answer at inference would turn both faithfulness judgment and option selection into near-trivial string matching, which is why the evaluation protocol withholds it. The task's difficulty is therefore not a matter of scaling or prompting a blind model harder, but of transferring what $a^\star$ would contribute into a model that never sees it at test time. This framing places the task in the learning-using-privileged-information setting (Vapnik and Vashist, 2009; Vapnik and Izmailov, 2015), which Lopez-Paz et al. (Lopez-Paz et al., 2016) unify with knowledge distillation (Hinton et al., 2015) as a transfer channel: soft teacher outputs carry exactly the information a privileged feature would otherwise supply. Most hallucination-mitigation systems instead keep a reference or a judge inside the inference loop (Manakul and Gales, 2023), which relaxes the blind constraint architecturally or pays repeated inference cost. Concurrent systems on this shared task (Ziani, 2026; Bekhouche et al., 2025; Swaileh et al., 2026; Bouchekif et al., 2025) report a similar detection-selection gap while keeping comparable machinery active at inference. We take the opposite position: keep the test-time system minimal, and confine every use of the gold answer $a^\star$ to training. Our system is a single low-rank adapter (Hu et al., 2021) over a frozen 7B Arabic instruction model (Bari et al., 2024), decoded greedily in two passes, with no reference-consuming module and no search. Privileged information enters only during training, through three mechanisms: detection and selection share one adapter, so option evidence regularizes the faithfulness decision; a teacher fine-tuned with $a^\star$ in its prompt supplies soft targets that expose calibrated uncertainty a one-bit label and a single letter cannot; and a privileged alignment term pulls the student's last-token representation toward a cached embedding of $a^\star$, so the gold answer shapes the representation space without ever entering a prompt. We further expand the 1,599 official training examples to 19,188 sequences through prompt paraphrasing and option permutation an augmentation that preserves blindness by construction. Our fine-tuned ALLaM-7B

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reaches 84.62 Comb on the official test set and 83.47 on dev, closing part of the blind-versus-gold gap with no change to the test-time architecture. Our contributions:

- We cast blind Arabic hallucination scoring as privileged-information transfer from a gold-answer-conditioned teacher.
- We combine gold-answer distillation with a contrastive privileged-information (CPI) term and blindness-preserving augmentation all training-time only, with inference unchanged.

2 Proposed Method

We propose a privileged-information distillation pipeline for Arabic hallucination scoring. A reference answer is available in training data and turns the problem into a comparison, but it is withheld at test time, leaving the model to judge a fluent answer on prior knowledge alone. Our method closes that gap by letting a teacher read the reference, then transferring its advantage into a student that never does, through three components: a privileged teacher fine-tuned with the reference in its prompt, which supplies targets richer than the one-bit label and single letter the annotations provide; knowledge distillation, which moves the teacher's output uncertainty to the blind student; and a contrastive privileged-information (CPI) term that additionally aligns their hidden representations, so the transfer is not confined to the final layer. Prompt and option-order augmentation, with teacher caches keyed to every augmented row, keeps this supervision dense rather than restricted to canonical phrasing. All three act only during training: inference is a single adapted model making two blind JSON calls, exactly as the competition protocol requires. Figure 1 contrasts the resulting objective with plain distillation. Each instance is a tuple $e = (q, \hat{y}, \mathcal{O}, y^{\text{lab}}, y^{\text{opt}}, a^{\star})$: an Arabic question q, a generated answer $\hat{y}$ to be judged, six candidates $\mathcal{O} = \{A, \ldots, F\}$, a hallucination label, the correct letter, and a reference answer $a^{\star}$. We decode two independent JSON views detection, $(q, \hat{y}) \rightarrow \{\texttt{predicted_label}\}$, and option selection, $(q, \hat{y}, \mathcal{O}) \rightarrow \{\texttt{selected_option}\}$ because the official score weights them separately, $\text{Comb} = 0.4\,\text{F1}_{\text{macro}} + 0.6\,\text{Acc}$; a single joint decode would let detection errors corrupt the more heavily weighted option accuracy. Detecting a hallucination is easy when $a^{\star}$ is visible and hard when it is not, yet the evaluation forbids showing it. We

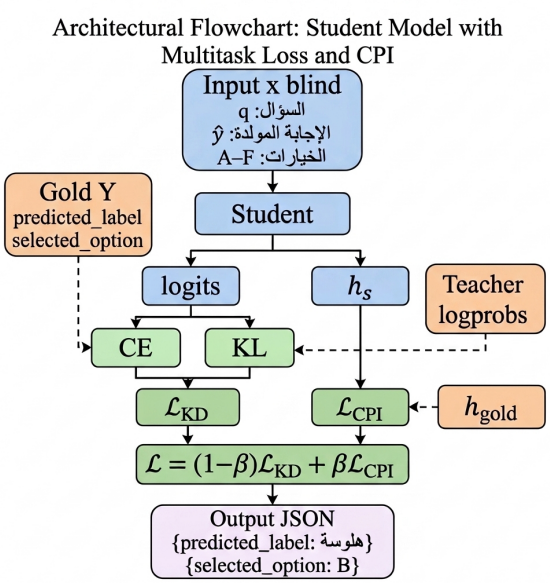


Figure 1: Overview of the Student training pipeline with Multitask Knowledge Distillation ($\mathcal{L}_{\text{KD}}$) and Contrastive Privileged Information Alignment ($\mathcal{L}_{\text{CPI}}$) for Arabic hallucination detection.

treat $a^{\star}$ as *privileged information*: available during training, absent at inference. We fine-tune a LoRA teacher $\mathcal{T}$ with cross-entropy on prompts that append the reference block containing $a^{\star}$, and never deploy it. One frozen forward pass records everything we need the top-K token log-probabilities $p_{\mathcal{T}}$ over the supervised span and the last-layer hidden state h_{gold} at the final supervised token so the teacher is paid for once and never re-run. The blind student $\mathcal{S}$ is initialized from $\mathcal{T}$: the two models share the output format and differ only in what the prompt reveals, so training starts from a policy that already emits valid JSON. Hard labels alone are a poor teacher here, since y^{lab} is one bit and y^{opt} one letter, both silent about *how close* a wrong option was. The teacher's soft distribution carries that information, so we distil it,

$$\mathcal{L}_{\text{KD}} = \alpha\, T^2\, \text{KL}\Big(p_{\mathcal{T}}^{(K)} \,\Big\|\, p_{\mathcal{S}}\Big) + (1-\alpha)\, \mathcal{L}_{\text{CE}}(Y, p_{\mathcal{S}}), \tag{1}$$

keeping cross-entropy on the gold JSON Y so the student stays anchored to the correct answer when the teacher is itself uncertain. Eq. 1 still supervises only the output layer, but the teacher's advantage lives earlier reading $a^{\star}$ changes how it *represents* the question, not merely which token it emits. We recover that with a contrastive term that pulls the blind representation toward the privileged one,

$$\mathcal{L}_{\text{CPI}} = 1 - \cos\big(h_{\text{student}}, h_{\text{gold}}\big), \tag{2}$$

evaluated at the same token position, and combine the two:

$$\mathcal{L} = (1 - \beta)\,\mathcal{L}_{\text{KD}} + \beta\,\mathcal{L}_{\text{CPI}}. \tag{3}$$

The terms are complementary rather than redundant Eq. 1 constrains *what the model says*, Eq. 2 constrains *what it encodes* before saying it so we keep β small (0.1): CPI is a regularizer on the hidden state, and a larger weight would trade away the token-level accuracy Comb actually measures.

3 Data and Experimental Setup

We evaluate on HalluScoring Subtask 2.2: 1,599 train, 800 dev, and 1,600 official test instances across four domains, with labels skewed roughly 59/41 toward *no_hallucination*. Augmentation expands the 3.2k-row training set to 19,188 rows through three prompt paraphrases $\times$ three option-order permutations. We compare ALLaM-7B, Qwen2.5-7B, and Falcon3-7B on augmented data. All models are trained with LoRA adapters for 5 epochs on augmented data , using fixed hyperparameters, with greedy JSON decoding throughout.

4 Results

We evaluate under the official protocol: two greedy JSON calls with no gold answer in the prompt, and $\mathrm{Comb} = 0.4\,\mathrm{F1}_{\text{macro}} + 0.6\,\mathrm{Acc}$. *Base* denotes unaugmented SFT on 3,198 rows; *Ours* denotes the full method (augmentation, distillation, and CPI). Table 1 reports both configurations for each backbone on the dev set and on the official test set; Table 2 isolates the contribution of each individual component on our submitted backbone.

Our submitted system, **ALLaM-7B**, improves over its own baseline on both splits: relative to Base, Ours yields a $+1.42$ Comb improvement on dev (82.05 to 83.47) and a $+1.46$ improvement on the official test set (83.16 to 84.62), with official performance exceeding dev the only backbone for which this holds. **Qwen2.5-7B** and **Falcon3-7B** exhibit a markedly different pattern: both show a modest dev-set gain over Base ($+0.23$ Comb for Qwen, $+0.22$ for Falcon), but this gain reverses on the official set, where Ours underperforms Base by 0.68 Comb for Qwen and by 1.83 Comb for Falcon the latter more than double the magnitude of the dev-set gain it is meant to reflect. This asymmetry indicates that the improvements from augmentation and distillation transfer reliably for the Arabic-native backbone but not uniformly across backbones.

Backbone	Method	Split	Joint	M-F1	Opt	Comb
ALLaM-7B	Base	Dev	71.12	82.12	82.00	82.05
		Official	72.69	82.36	83.69	83.16
	Ours	Dev	72.75	82.11	84.38	**83.47**
		Official	74.25	82.27	86.19	**84.62**
Qwen2.5-7B	Base	Dev	73.50	82.79	83.50	83.22
		Official	69.56	78.90	82.94	**81.32**
	Ours	Dev	74.88	84.13	83.00	**83.45**
		Official	69.31	77.85	82.50	80.64
Falcon3-7B	Base	Dev	62.12	79.04	72.50	75.12
		Official	61.94	76.63	74.62	**75.43**
	Ours	Dev	62.62	78.67	73.12	**75.34**
		Official	60.25	74.39	73.06	73.60

Table 1: Base versus Ours (full method) on the dev set and the official test set, across three backbones. ALLaM-7B is our submitted system. Bold marks the better Comb within each backbone–split pair.

Table 2 isolates the contribution of each training-time component on our submitted backbone. On dev, the terms combine additively: Base CE attains 82.05 Comb, augmentation raises this to 82.89, distillation improves it further to 83.35, and CPI contributes a final $+0.12$ to reach 83.47. CPI's benefit also holds in the absence of distillation on dev, where Aug CE+CPI attains 83.35 Comb, a $+0.46$ improvement over Aug CE alone; this effect does not, however, carry over to the official set, where Aug CE+CPI (84.01) falls 0.04 Comb *below* Aug CE alone (84.05), a sign reversal relative to dev. CPI's contribution on top of distillation is the one effect that retains its sign across both splits ($+0.12$ on dev, $+0.65$ on official), and the resulting configuration, Aug KD+CPI (**Ours**), remains the strongest overall at 84.62 Comb. We conclude that CPI constitutes a reliable addition on top of distillation but not a substitute for it: its apparent benefit over cross-entropy alone appears to be a dev-set artifact that does not generalize. Across configurations, gains concentrate almost entirely in option accuracy, while Macro-F1 remains within a narrow 81-83 range throughout consistent with the hypothesis that the withheld gold answer $a^\star$ primarily disambiguates *which* option is correct, rather than *whether* the given answer is faithful.

5 Discussion

Two results anchor our analysis: CPI's contribution is well-characterized on ALLaM-7B, and the backbone ranking in Table 1 is consistent with each

Config	Joint	M-F1	Opt	Comb
Base CE (Dev)	71.12	82.12	82.00	82.05
Base CE (Off.)	72.69	82.36	83.69	83.16
Base KD (Dev)	70.38	81.07	81.25	81.18
Base KD (Off.)	72.00	82.02	83.38	82.83
Aug CE (Dev)	72.88	82.16	83.38	82.89
Aug CE (Off.)	73.88	82.35	85.19	84.05
Aug KD (Dev)	72.62	82.55	83.88	83.35
Aug KD (Off.)	73.38	82.04	85.25	83.97
Aug CE+CPI (Dev)	73.50	82.76	83.75	83.35
Aug CE+CPI (Off.)	73.81	82.15	85.25	84.01
Ours (Dev)	72.75	82.11	84.38	83.47
Ours (Official)	**74.25**	**82.27**	**86.19**	**84.62**

Table 2: Ablation on our submitted system (ALLaM-7B): dev vs. official test.

model's documented Arabic-language pretraining. CPI (Eq. 2) pulls the student's blind hidden state toward a frozen, gold-conditioned reference, constraining the representation a decision is read *from* rather than the output distribution itself as $\mathcal{L}_{\text{KD}}$ does. Table 2 supports this distinction directly: CPI's gain on top of distillation is the only term that retains its sign across both splits ($+0.12$ dev, $+0.65$ official), while distillation alone does not Aug KD trails Aug CE on official (83.97 vs. 84.05), reversing its dev-set lead. The gain is concentrated almost entirely in option accuracy ($+0.50$ dev, $+0.94$ official), with Macro-F1 nearly unmoved. This is consistent with where the two objectives act: detecting a hallucination is often resolvable from surface-level inconsistency, while selecting the correct option among six candidates requires encoding *which* one is correct information a representation-level signal is better positioned to carry than an output-level one. The per-backbone ranking in Table 1 matches each model's documented Arabic pretraining depth. ALLaM-7B has a dedicated 1.2-trillion-token Arabic/English pretraining stage (Bari et al., 2024). Qwen2.5-7B includes Arabic as one of roughly 30 supported languages, with no stage specific to it (Qwen et al., 2025). Falcon3-7B, the backbone we evaluate, had no native Arabic tokenizer or embedding support at all: TII's own release notes for a subsequent Arabic-adapted variant confirm the base checkpoint required extending the vocabulary with 32,000 Arabic-specific tokens to add Arabic support after the fact (TII Team, 2025). This ordering ALLaM, then Qwen, then Falcon matches the Comb ranking in Table 1, and offers a plausible account for why the same recipe yields a stronger model on ALLaM. It does not explain the dev-to-official reversal visible for Qwen2.5-7B and Falcon3-7B in the same table: corpus composition is a property of the backbone alone, while the reversal is a difference in generalization between splits, and we lack the per-backbone ablation needed to attribute it to a specific component.

6 Conclusion

We presented a privileged-information distillation pipeline for Arabic hallucination scoring: a teacher reads the reference answer and never deploys it, transferring its advantage to a blind student through output-level distillation and a contrastive term aligning hidden representations. Our fine-tuned ALLaM-7B reaches 84.62 Comb on the official test set and 83.47 on the dev set. CPI improves every backbone we tested, on both splits, by concentrating its gain in answer selection rather than detection evidence that aligning representations, not just output distributions, recovers the teacher's advantage reliably.

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